rubber..

All You Need to Know

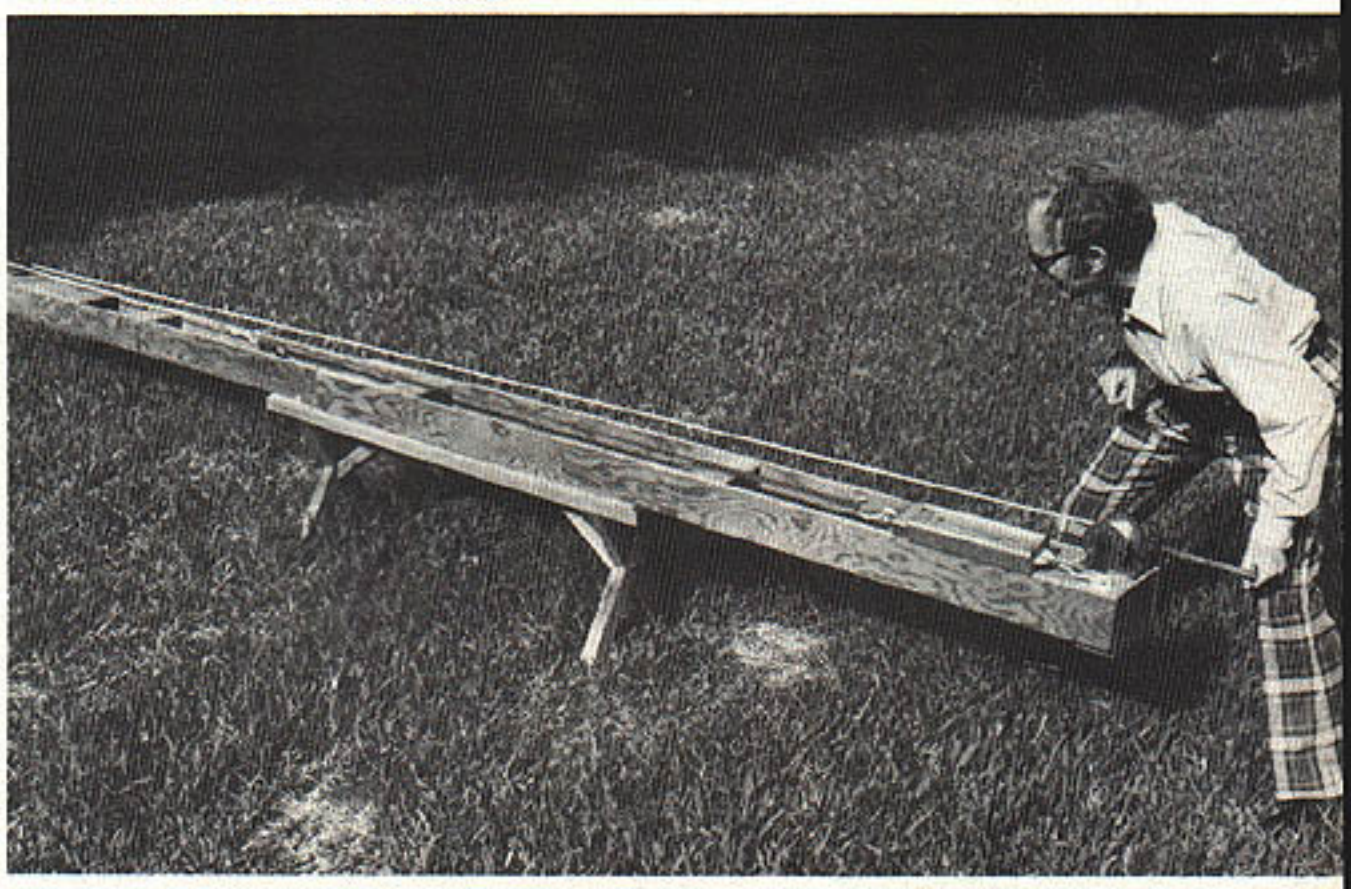
The first of a two-part article covers the testing program and its results. How to handle rubber for optimum results will be next month's subject. ■ Fred Pearce

The author launching his '74 Wakefield. It won first at both the 1974 and 1975 Nationals.

SINCE 1973 good rubber strip has been a problem to find. The trouble started when the chief source, the Pirelli factory near Milan, Italy was reorganized. It was split off from the Pirelli company and became part of a new company called Filati. The Pirelli company still makes the sheet rubber, which is sold to Filati who strips it to size and markets it.

Beginning with the 1973 batch the Filati tested by the writer varied widely in energy storage and appearance. The old fashioned product was brown and had energy storage around 3300 foot pounds per pound. In 1973 a test of a very black Filati gave only 2250 foot pounds per pound. In 1974 a gray batch tested at an encouraging 3310, but in 1976 a batch gave 2600, and in 1977, a batch gave only 1500 foot pounds per

Editor's Note: *Mr. Pearce's time consuming research–another example of scientific reporting* **MA** *occasionally does– is a bonanza for all people who fly rubber events. Encouragement is extended to "thinkers" in the vast RC field–and to others in both CL and FF.*



What could be the world's largest slingshot is really Fred's special test rig on which he is testing a Wakefield motor. A 200-pound scale measures the pull force. Stretch distance is calibrated on the edge of the frame. In any category one finds dedicated researchers—fortunately.

TABLE A
RUBBER TEST RESULTS
stretch method at 75 degrees F

Vintage	Type	Source	Comment	Test Date	Elongation	Energy ft. lb. / lb.
1958	Pirelli		Brown, ribbed	Jan 64	7.5	2940
1962	Pirelli			Oct 63	8.0	2980
1963	Gitlow	Gitlow	Gray, smooth	Jan 64	6.0	2890
1963	Pirelli	Tenny		Jan 64	7.4	3300
1964	Pirelli	Cannizzo		1964	8.1	3525
1968	Pirelli	Zaic		May 70	7.6	3100
1968	Pirelli	Zaic	Repeat test, brittle	Aug 75	6.9	2950
1969	Pirelli	Sig		Jun 71	7.8	3160
1970	Pirelli	Zaic		May 72	8.3	3440
1970	Pirelli	Zaic	Repeat test, brittle	Aug 74	7.9	3310
1971	Pirelli	F and D		Aug 73	8.0	3290
1972	Pirelli	Zaic		Jul 73	8.2	3310
1972	Pirelli	Zaic	Repeat test, brittle	Jul 77	7.3	3540
1973	Filati	F and D	Black	Apr 73	7.7	2250
1973	Sig	Sig	Gray, smooth	Jul 73	8.4	3140
1973	FAI	FAI	Gray, smooth	Jul 73	7.8	2500
1974	Pirelli	Old Timer	Vintage date?	Aug 75	7.8	3210
1974	Pirelli	Old Timer	Repeat test, brittle	Jul 77	7.9	3600
1974	Filati	Zaic	Gray	Oct 74	8.2	3310
1976	Vintage Aero	Vintage Aero	Gray, smooth, soft	July 76	9.0	3200
1976	Vintage Aero	Vintage Aero	Repeat test	May 77	8.3	3060
1976	Filati	Aeromodels	Black	Apr 76	8.1	2600
1976	Japanese	Tsuda	Brown, smooth	May 77	6.6	2620
1976	FAI	FAI	Feb. 76 batch	May 77	7.3	3400
1976	FAI	FAI	Aug. 76 batch	Sep 76	8.0	2980
1976	FAI	FAI	Dec. 76 batch	May 77	7.7	3380
1977	FAI	FAI	Mar. 77 batch	Jun 77	7.8	3450
1977	FAI	FAI	Jun 77 batch	Jul 77	8.0	3320
1977	Filati	Filati	Dark brown, opaque	Mar 78	6.4	1570
1978	FAI	FAI	Apr 78 (spool 1)	Sep 78	8.0	3440
1978	FAI	FAI	Apr 78 (spool 2)	Sep 78	8.25	2970
1978	Pirelli	Cannizzo	Translucent yellow brown	Sep 78	8.1	3725

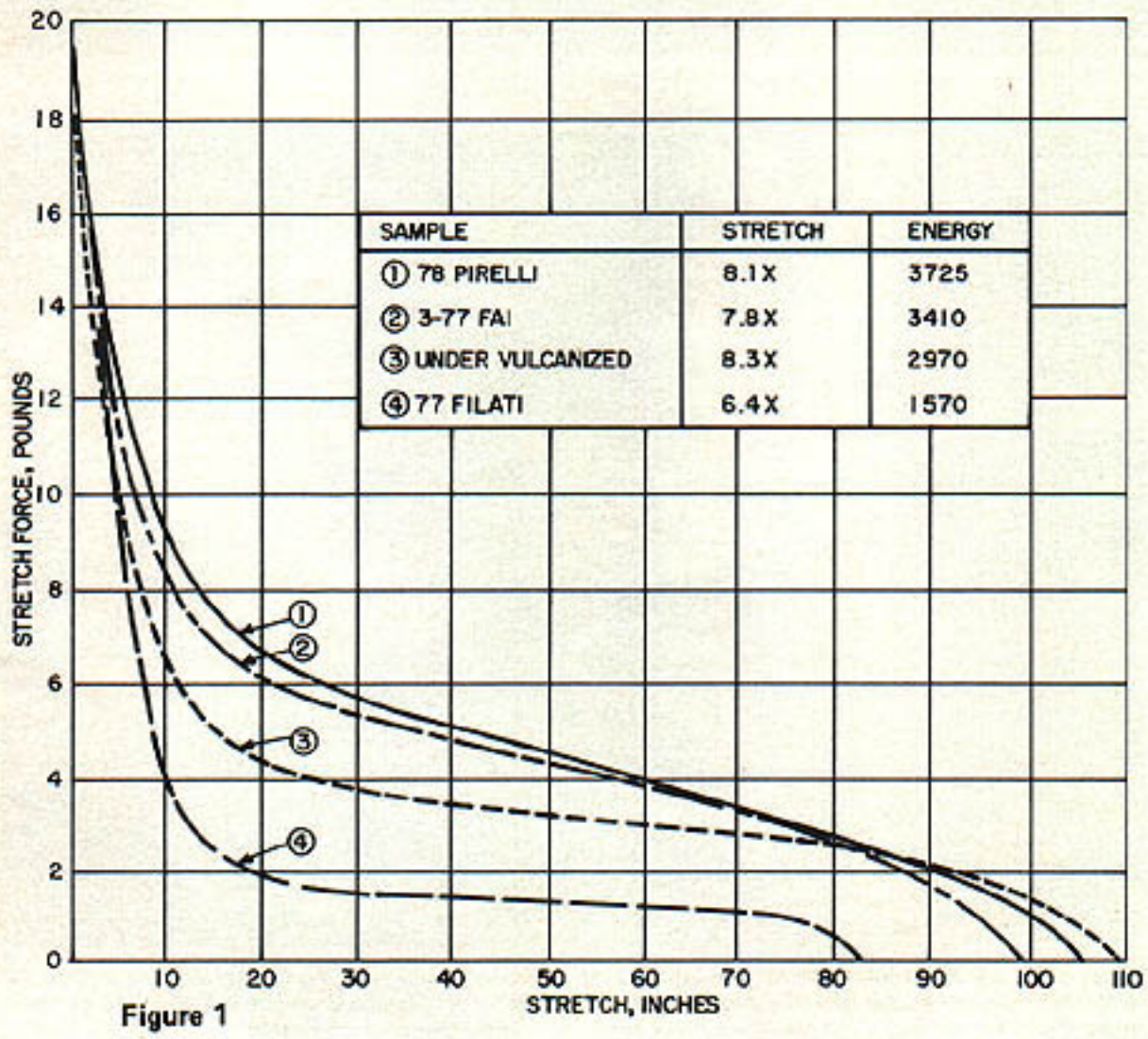


Figure 1

pound. These test findings and the disappointing performance experienced by models fliers when using Filat led to the seeking out of a new source.

The problem appears to be in the understanding of what a model flier calls quality. The various manufacturers did not get our message for a rubber which stores energy. The new supplies of rubber coming forth ranged widely in quality, lately reaching the level of pre-1973 Pirelli. But a change has now taken place. Excellent quality rubber is available again. To quote a rubber flier of the old school, Mike Dikovitzky, "The fun has been put back into flying rubber powered models!"

Rubber is a complex mixture of many ingredients which must be mixed and processed just right in order to produce a product with optimum characteristics for a given use. Small changes in the ingredients or in the mixing sequence, or in the vulcanization process, can produce wide changes in physical properties. When the supply of Pirelli became irregular, initially, no other manufacturer rose to the occasion to fill the gap. This is understandable, since there is a very small demand for model aircraft rubber strip. A consistent supply of high quality strip, as judged by model fliers, only became available after Ed Dolby had the perseverance to work with a manufacturer over a protracted period of time. In response to Ed's specifications, the manufacturer produced batches of rubber with characteristics that he thought would satisfy model builders. Through several reiterations, a recipe was developed that, when tested by model builders, gave good results. Once the manufacturer understood how to correlate his production with good performance in a model airplane, he optimized his recipe to produce a strip of consistent high quality.

In mid-1978 Pirelli again entered the picture. After a five year lapse, they are producing a product referred to as "Special Pirelli for Model Fliers," which is compounded without carbon black. It is yellow brown and almost translucent. It has the ribbed texture of the old fashioned product. This new rubber gives higher energy storage than any rubber encountered by the author. The storage runs above 3700 foot pounds per pound at 75° F. In addition to FAI and the new Pirelli, we have several other sources of rubber, including two which the author has not tested recently. Sig and Vintage Aero are offering rubber which is reported to give very good results.

Manufacture of Strip Rubber: Strip rubber is manufactured from natural crude rubber, synthetic rubber, stearic acid, zinc oxide, sulphur, carbon black and small amounts of catalysts and conditioners—such as Altax, Neozone D, Accelerator 808, Titanox, etc. The best quality rubber uses all virgin stock, as reclaimed-rubber imports have poor characteristics. A gray

or black color is imparted by carbon black. Brown rubber contains little or no carbon black. The recipes for rubber manufacture are almost always proprietary, since there is an element of witchcraft in obtaining desired characteristics. A guess at the recipe for the current smooth gray American manufactured rubber is:

Ingredients	Parts by Weight
Pale Virgin Crepe (natural rubber)	85
Synthetic rubber	15
Stearic Acid	1
Zinc Oxide	5
Altax (Accelerator Catalysts)	1
Neozone D (Activator Catalysts)	1
Sulphur	3
Carbon Black	1 (or less)
	112

It is presumed that the product would be improved by the deletion of the synthetic rubber and the carbon black. But, this is a presumption and remains to be determined. In a mill, the ingredients are mixed at about 120° F in the following order:

Step	Ingredients	Mixing time
1	Natural Rubber and synthetic rubber	4 minutes
2	Stearic Acid (softens mixture)	2 minutes
3	Zinc oxide, Altax, Neozone D and Carbon Black	2.5 minutes
4	Sulphur (acts to vulcanize during curing process)	1.5 minutes

After the mixing the rubber is run through roller to produce a sheet of the desired thickness, typically .043".

The sheet comes out five to six feet wide and about 900 feet long. It is then dusted with talcum powder to keep it from sticking to itself. After rolling it up, it is put into an electronically controlled vulcanizing oven where it is subjected to steam at about 285° F for around 30 minutes. The degree of vulcanization has a very dramatic effect on the tensile properties. Vulcanization is the cross linking of the polymer chains and is brought about by the action of the sulphur at high temperature. After removal from the oven, the rubber is allowed to cool to room temperature prior to the slitting process.

The first step in slitting the rubber is to cut it into belts 7.5 inches wide. These are sliced end to end with a tapered and glued joint. The total length now of the continuous 7.5 inches belt is about 9000 feet. The next step is to run the belt through a slitting mill set for the desired widths which are typically 1/16, 3/32, ⅛, 3/16 and ¼ inch strips. After passing through the slitting, the strip is run off on spools containing about a pound of rubber. This amounts to 200 feet for ¼ inch strip. A typical batch weight is about 1500 pounds.

The new Pirelli is about the same size as the old Pirelli. A 38.8 gram dry weight Wakefield motor makes up at 16 strands by about 16½ inches long. The FAI rubber is a little thicker, with 14 strands being about the same length as the Pirelli and takes about the same turns (320). The Sig and Vintage Aero rubber is generally thinner than Pirelli and require more strands to get the equivalent cross-section area. Rather than strands it is better to think of equivalency in the skein length of made up motors.

See Table A for a tabulation of rubber test results since 1958. A good Wakefield should be able to do over 180 seconds duration with rubber with over 3100 foot pounds per pound energy storage. Likewise, a very good Coupe should do over 120 seconds with 3100 rubber. See. Fig. 1 for a comparison of the power curves of Pirelli, Filati and FAI Rubber.

Rubber testing: Rubber stores energy in tension which can be produced by either stretching it or winding it. Tests by either method give comparable results. The writer has compared his stretch tests over a long period with the winding tests of Chris Matsuno. Our tests are in close agreement.

Fig. 2 compares results using the two methods which shows the differences in the power curves due to the characteristic difference in behavior of rubber when being wound or stretched. Excluding the strongly non-linear portions of the beginning and end of the curves, the torque for a wound sample varies proportionally to the 1.5 power of the number of strands, whereas, the pull force for stretched rubber varies directly proportionally to the number of strands. Both test samples were of the same weight. Both methods have the same area beneath the curve which indicates identical energy storage. The areas are the products of force times distance, and turns times torque, respectively and represent foot pounds of energy. These curves demonstrate identical results in using either method to determine energy storage.

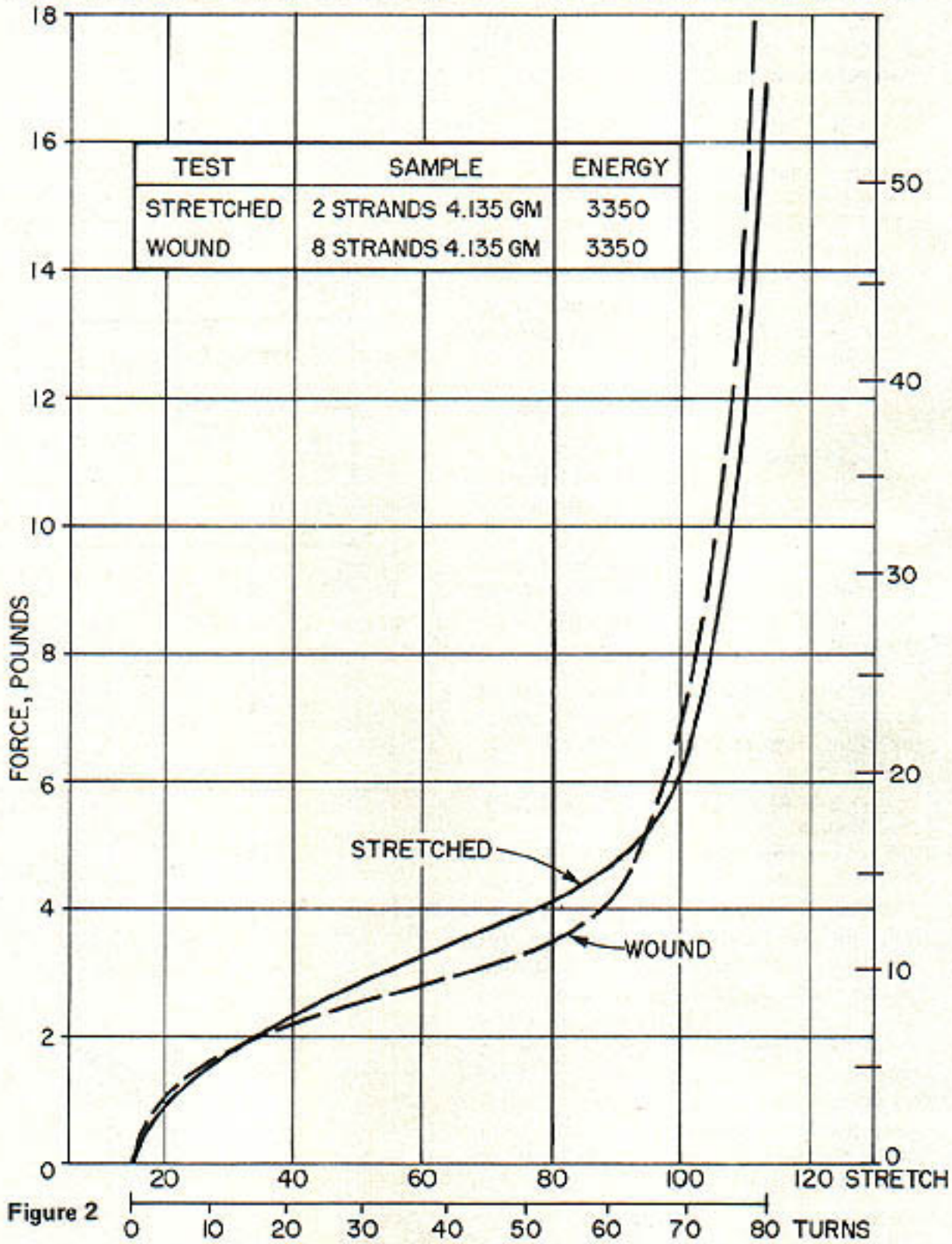


Figure 2

It is much more difficult to wind a motor to ultimate capacity with accuracy and repeatability than to stretch it to its maxi-

mum length. Winding energy storage is affected by the distance stretched as you put in the turns, as well as by the percentage of turns put in at various distances. Hence, I standardized on stretch testing in order to obtain consistency.

To obtain comparable results from test to test, I felt that it was necessary to pull all samples to the same ultimate tensile stress very close to the breaking point. Figuring the original cross-section and pulling with a proportional force does not do the job, because the more stretchy rubber would be pulled to a higher tensile stress. This problem was solved by pulling a motor in two stages. The first pull is computed with the following formula:

$$\text{Force}_1\ (\text{pound}) = \frac{45 \times \text{weight (grams)}}{\text{Original Length (inches)}}\ *$$

This force is strong enough to pull the motors to near their maximum stretch. Then a second pull force is computed to the following formula:

$$\text{Force}_2\ (\text{pounds}) = \frac{430 \times \text{weight (grams)}}{\text{Stretched Length from } F_1 \text{ (inches)}}\ *$$

**The purely metric versions of these formulas have constants of 52 and 496 respectively with F in kilograms, W in grams and L in centimeters.*

Applying F_2 will stretch the motors an additional small amount to near the breaking point. For example, the March 1977 FAI Rubber sample of two strands of ¼ inch was pulled to F_1 = 15.3 and F_2 = 19.2 pounds.

The question arises, would stretchier rubber be pulled to a higher tensile stress and thus store more energy, therefore, reducing the validity of comparing different samples? This problem is mitigated by the shape of the energy storage curve (see Fig. 1). As you pull the rubber to very high tensile stress, the change in elongation becomes less and less, with the final result becoming a climbing-the-wall situation where you get almost no change in stretched length as you approach breaking stress. Thus, there is very little additional energy stored, since the incremental product of force times distance is very small.

My rubber testing machine is shown in the photograph. It has a 200-pound scale for handling full-size Wakefield motors and a 20-pound scale for testing single loops of rubber. A boat winch is provided for stretching the samples along the calibrated 150-in. frame. Test motors are attached to a hook on the scale and to a hook on the end of a rope which travels around a pulley on the far end of the frame and back to the winch.

In use my testing procedure is as follows:

(1) Weigh a sample loop suitable for the 150-in. frame, say between 12 and 16 inches long. Subtract out the weight of the knot and tag ends.

(2) Lubricate sample with green soap and glycerine type of rubber lubricant. Don't get the knot soaked with lube, as the high stress will cause it to slip. I put extra talcum powder on the ends before tying a square knot with a single granny knot on top. Saliva helps keep the rubber from tearing while pulling the knot tight.

(3) Compute pull force F_1 and break in rubber by pulling to this force for 5 minutes. Wait at least one hour before next step.

(4) Using the stretched length from the first pull, compute the second pull force F_2.

(5) Pull the motor to F_2 taking care to take about 15 seconds to pull the last 10% of the desired force.

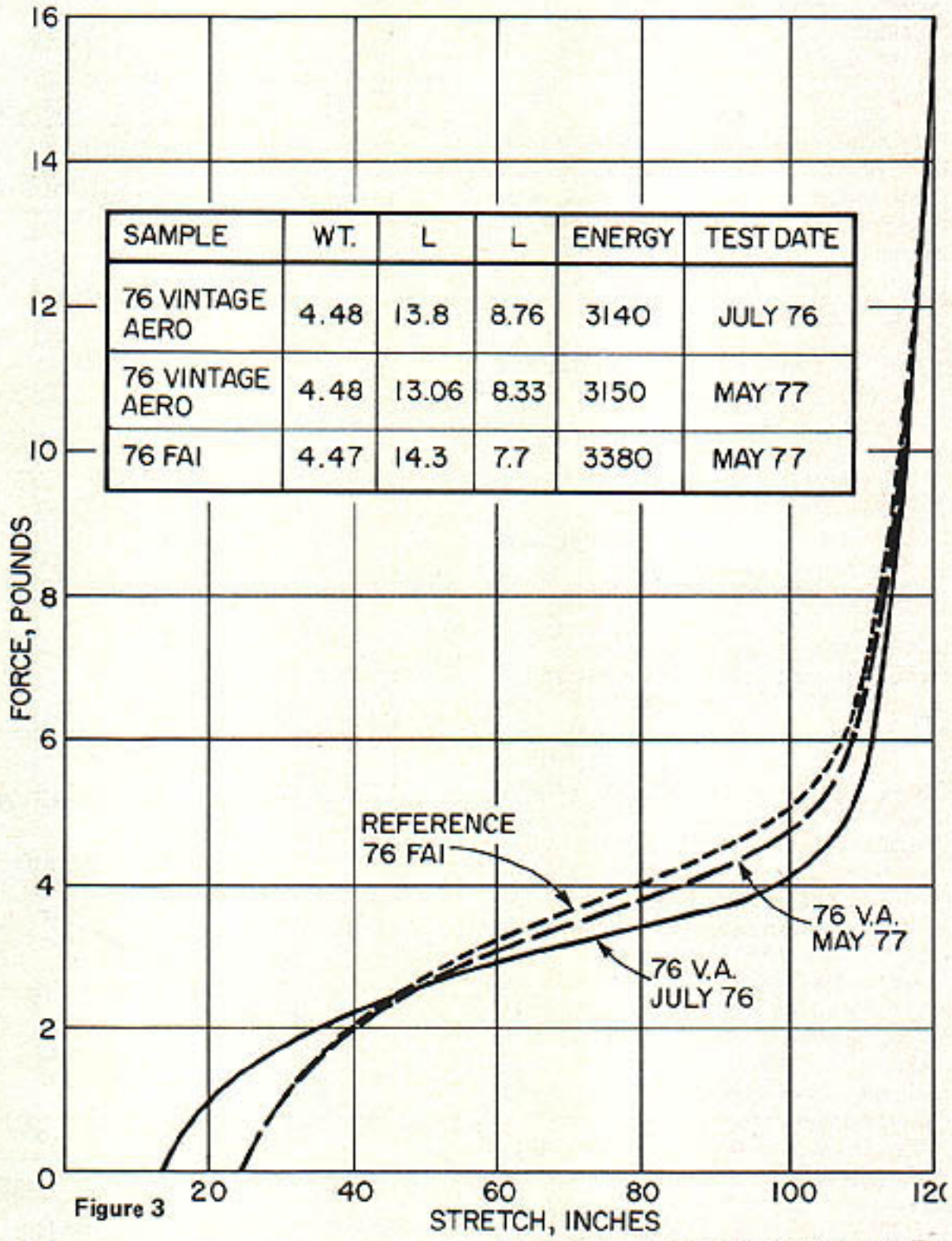
(6) Immediately read and record the force and stretch distance. Backing off in small increments, step down in about a dozen steps to zero tension recording each step.

(7) Correct the stretch distances for any deflection of your force scale.

(8) Plot the force and stretch distance on graph paper. Measure the area beneath the curve. Typically, I integrate the sum of about a dozen steps along the stretch scale.

(9) The area beneath the curve is in foot pounds and represents the energy. When divided by the sample weight, one gets the energy storage in foot pounds per pound. The writer believes his apparatus gives a result to 1% accuracy.

Foot pounds per pound reduces simply to feet. The interpretation of this is that a sample of rubber could lift itself vertically a distance equal to its energy storage. For example, a sample of 3300 could lift itself 3300 feet. A model airplane with 25% of its weight in rubber could climb .25 × 3300 = 825 feet if all the energy were used at 100% efficiency.



SAMPLE	WT.	L	L	ENERGY	TEST DATE
76 VINTAGE AERO	4.48	13.8	8.76	3140	JULY 76
76 VINTAGE AERO	4.48	13.06	8.33	3150	MAY 77
76 FAI	4.47	14.3	7.7	3380	MAY 77

Figure 3

Standards for Rubber Strip: What are the standards for good rubber? In the past when good Pirelli was available, good quality gave energy around 3300 foot pounds per pound and stretched at maximum to about 8 times its original length.

Rubber below 3000 was considered definitely below par. Rubber which stretched to less than about 7.3 times its original length was looked upon with suspicion as it was thought to be getting old. Rubber that stretched over 8.2 times is thought to be soft and probably under-vulcanized.

Some values for rubber samples are given in Table A.

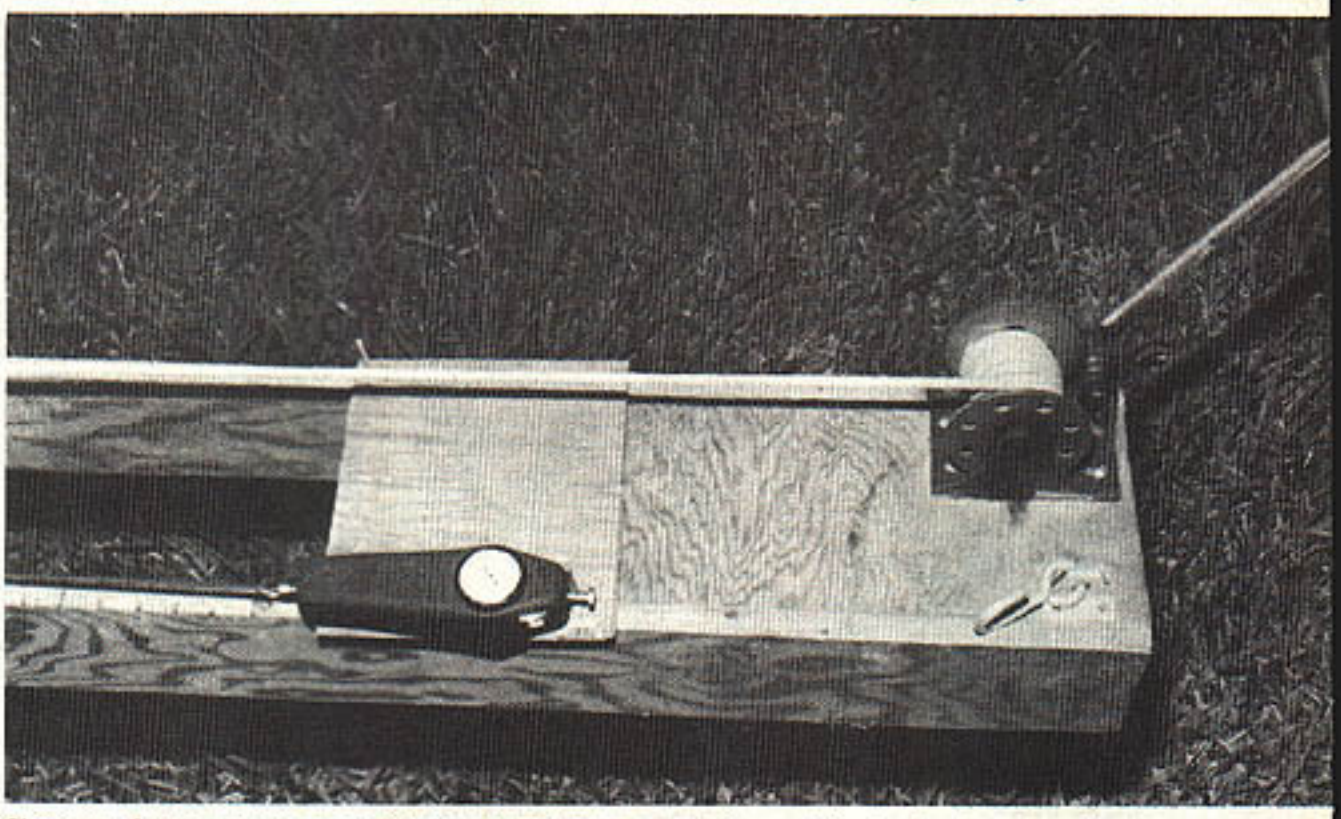
Rubber which is below par in energy storage typically has a power curve such as the two lower curves in Fig. 1. This shape power curve can give trouble in use because the sharp spike of power is not utilized effectively. The model is inclined too much after the first few seconds because the torque is falling more rapidly than the airplane can respond. This effect is especially noticeable when there is ground turbulence, and when the model is climbing very steeply where it is easily stalled or upset. Rubber which is too soft, but of good energy storage such as the lower curve of Fig. 3, also gives this initial poor climb characteristic.

Aging Effects: Rubber is a polymer. Aging results in the continuation of uncompleted vulcanization, reactions usually resulting in chain-lengthening and in increased crosslinking between chains. These processes lead to toughening of the rubber. Aging of rubber beyond optimum condition results in chain scission (decomposition), resulting in shorter chains combined with increasing crosslinking of an excessive nature. The product becomes increasingly brittle. The balance of the chain lengthening, chain decomposition and crosslinking can vary widely. Generally, good quality rubber will slowly lose its ability to stretch while even more slowly losing its energy storage capability. Rubber becomes tougher with age, eventually becoming brittle. Brittle rubber tears at knots and slight imperfections. It also is inclined to explode, breaking all strands in a skein as it is wound.

Fig. 3 gives an example of an under-vulcanized rubber which became tougher with a year of age. Its power curve initially had the sharp peak torque characteristic whose undesirable characteristics were described previously under "Standards for Rubber Strip." Age helped the shape of the power curve while slightly lowering the energy storage. An occasional example of rubber will increase its energy storage with age, but this is uncommon and is associated with becoming dangerously brittle.

Note in Table A the generally declining elongation with age of the 1968 and 1972 Pirelli. By 1975 the 1965, 1966, 1967, 1969 and 1970 Pirelli were all too brittle to use, although the 1968 was useable but dangerous. By 1976 the 1972 Pirelli had become dangerously brittle although its energy was excellent; note that its elongation has declined from 8.2 initially to 7.3 today.

A second article will appear next month on the handling of the new generation rubber. In addition to the conventional questions of making up skeins, breaking them in and winding to maximum turns, two additional topics will be covered which are the effect of temperature on energy storage and the effect of holding a fully wound motor on energy storage.



Close-up of test rig with special 20-pound precision scale being used for testing two-strand motors. Note the boat winch employed for pulling the rubber motors undergoing testing.

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